

Questions to be answered individually

Read the paper “Smartphone Ownership, Age of Smartphone Acquisition, and Health Outcomes in Early Adolescence” by Barzilay et al., *Pediatrics*, 2026 and answer the following questions.

Both the paper and question sheet are available in Canvas under the assignment “Homework: Science and Data”.

Actively use AI to get help if there is something you do not understand.

Part 1: Understanding the Research Question

1. What is the primary research question of the paper?
2. What are the main exposures (independent variables) studied?
3. What are the health outcomes (dependent variables)?
4. What is an odds ratio? What does it mean if the odds ration is larger than 1?

Part 2: The study design

1. Did the researchers design a study and collect data to answer a research question?

- ### Part 3: Testing differences in groups

2. The two-samples t -test used by the authors is defined as

Can you identify in the table the numbers that are used to compute the test statistics?

3. What does the reported p-value say about the tested hypotheses?

4. Look at the effect size, what is the difference between the two groups in terms of sleep hours? Do you think it is a big difference?

5. Why do you think that it is often stressed that scientists should look at effect sizes and not only p-values? How can a very large sample size make small effects statistically significant?